

FirstKnock Bug Report — Route Generation & Square Draw Crashes

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Route Generation & Square Draw Crashes

2026-04-08

2times P0 Crash Bugs Fixed	1times P1 Crash Bug Fixed	1times Architecture Verified	2 Files Affected
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Table 1 — Bug Report Metadata

Field	Detail
Date	2026-04-08
Severity Summary	2times P0 crash bugs fixed, 1times P1 crash bug fixed, 1times architecture verification
Files Affected	routeOptimizer.ts, MapDrawTool.jsx
Status	Fixes Applied — Pending QA Sign-Off

Figure 1 — FirstKnock Route 1 displayed on map (Dallas, TX). Yellow lines show the boustrophedon mail-carrier walking path across 84 stops.

Bug 1: Route Generation Crash

Severity: P0 — Complete Crash. Clicking "GENERATEROUTES" in the Route Builder throws a TypeError at runtime. Route generation is completely non-functional. No route is displayed on the map.

Symptom

Clicking "GENERATEROUTES" in the Route Builder throws a TypeError at runtime. Route generation is completely non-functional. No route is displayed on the map.

Root Cause Summary

Five distinct crash sites were identified in routeOptimizer.ts, all stemming from the same class of error: array-as-scalar access and missing empty-array guards. Each site is documented below with before/after code.

Table 2 — Bug 1 Root Cause Index

Root Cause ID	Location	Error Class	Impact
1a	fatigueAwareFrontLoad()	Missing empty-array guard	Crash on route[0] when array is empty
1b	returnToStart block	Array accessed as scalar	calculateDistance() receives NaN, throws
1c	distanceFromStart block	Array accessed as scalar	calculateDistance() receives NaN, throws
1d	generateAppleMapsUrl()	Array assigned to origin variable	origin.lat is undefined, URL builder crashes
1e	mailCarrierOrder()	No lat/lng validation filter	NaN coordinates corrupt sort order, crash downstream

Root Cause 1b — returnToStart Block: orderedProps Used as Scalar

orderedProps is typed as Property[] (an array). The returnToStart distance calculation accessed orderedProps.lat and orderedProps.lng directly — both are undefined on an array — causing calculateDistance() to receive NaN arguments and throw.

BEFORE (broken):

```
const returnDist = calculateDistance(orderedProps[orderedProps.length - 1].lat,
orderedProps[orderedProps.length - 1].lng,orderedProps.lat, // leftarrow undefined —
orderedProps is Property[], not PropertyorderedProps.lng, // leftarrow undefined);
```

AFTER (fixed):

```
const returnDist = calculateDistance(orderedProps[orderedProps.length - 1].lat,
orderedProps[orderedProps.length - 1].lng,orderedProps[0].lat, // first element of the array
orderedProps[0].lng,);
```

Root Cause 1c — distanceFromStart Block: Same Array-as-Scalar Bug

Same pattern in the distanceFromStart calculation immediately below the returnToStart block.

BEFORE (broken):

```
distanceFromStart = calculateDistance(startLocation.lat, startLocation.lng,orderedProps.lat, //
leftarrow undefinedorderedProps.lng, // leftarrow undefined);
```

AFTER (fixed):

```
distanceFromStart = calculateDistance(startLocation.lat, startLocation.lng,orderedProps[0].lat, //
```

```
orderedProps[0].lng,    );
```

Root Cause 1d — generateAppleMapUrl: properties Array Assigned to origin

const origin = properties assigned the full Property[] array to origin. The subsequent template literal `\${origin.lat},\${origin.lng}` accessed undefined on the array, crashing the URL builder.

BEFORE (broken):

```
const origin = properties; // leftarrow entire array, not a single Propertyconst originStr = `${origin.lat},${origin.lng}`; // crashes: origin.lat is undefined
```

AFTER (fixed):

```
const origin = properties[0]; // first property in the routeconst originStr = `${origin.lat},${origin.lng}`;
```

Root Cause 1a — fatigueAwareFrontLoad: No Empty-Array Guard

When mailCarrierOrder returns [] (e.g., all properties have invalid coordinates), fatigueAwareFrontLoad immediately called routeDist(route) which accessed route[0] on an empty array — undefined — and crashed.

BEFORE (broken):

```
function fatigueAwareFrontLoad(route: Property[]): Property[] {if (route.length <= FATIGUE_FRONT_LOAD_STOPS)return route; // passes for length=0const routeDist = (r: Property[]) => { ... };const baselineDist = routeDist(route); // route[0] is undefined rightrightarrow crash
```

AFTER (fixed):

```
function fatigueAwareFrontLoad(route: Property[]): Property[] {// BUG FIX: guard against empty array before any array accessif (!route || route.length === 0) return route ?? [];if (route.length <= FATIGUE_FRONT_LOAD_STOPS)return route;...
```

Root Cause 1e — mailCarrierOrder: No Filter for Undefined lat/lng

Properties with undefined or NaN lat/lng values were passed through to boustrophedonStreet, where calculateDistanceFast received NaN arguments, silently corrupting the sort order and potentially crashing downstream consumers.

BEFORE (broken):

```
export function mailCarrierOrder(properties: Property[], ...): Property[] {if (!properties || properties.length === 0) return []; // No coordinate validation — undefined lat/lng passed to boustrophedonStreetconst streetGroups = groupByStreet(properties);
```

AFTER (fixed):

```
export function mailCarrierOrder(properties: Property[], ...): Property[] {if (!properties || properties.length === 0) return []; // BUG FIX: filter out properties with missing/NaN coordinates
```

```
const validProperties = properties.filter(p => p && typeof p.lat === 'number' && !isNaN(p.lat) &&typeof p.lng
=== 'number' && !isNaN(p.lng));if (validProperties.length === 0) return [];const streetGroups =
groupByStreet(validProperties);
```

Files Affected	routeOptimizer.ts
Severity	P0 — Complete crash, route generation non-functional

Bug 2: Square Drawing Tool Crash

Severity: P1 — Feature Non-Functional. Selecting the square draw tool and tapping the map throws a `TypeError` or produces an invalid polygon that crashes downstream `turf.js` calls. The square draw mode is completely non-functional.

Symptom

Selecting the square draw tool and tapping the map throws a `TypeError` or produces an invalid polygon that crashes downstream `turf.js` calls. The square draw mode is completely non-functional.

Root Cause Summary

Three candidate root causes were identified, all confirmed present in `MapDrawTool.jsx`.

Table 3 — Bug 2 Root Cause Index

Root Cause ID	Description	Impact
A	Missing explicit 'square' case in draw handler	<code>clickHandler</code> is undefined <code>rightarrow</code> <code>TypeError</code> on
B	Square polygon not Mercator-corrected	<code>map.addListener</code> Visually distorted rectangle at non-equatorial latitudes; <code>turf.area()</code> returns incorrect values
C	Square polygon ring not closed (missing 5th point)	<code>turf.booleanPointInPolygon</code> throws "ring is not closed" on every property-in-square check

Root Cause A — Missing Explicit square Case in Draw Handler

The draw handler had explicit branches for 'circle' and 'polygon' but no 'square' case. The square

mode fell through to undefined, and the runtime attempted to call undefined().

BEFORE (broken):

```
// Only circle and polygon handled — square falls throughif (drawMode === 'circle') {clickHandler = handleCircleClick;} else if (drawMode === 'polygon') {clickHandler = handlePolygonClick;}// square: clickHandler is undefined rightarrow map.addListener('click', undefined) rightarrow TypeError
```

AFTER (fixed):

```
if (drawMode === 'circle') {clickHandler = handleCircleClick;} else if (drawMode === 'polygon') {clickHandler = handlePolygonClick;} else if (drawMode === 'square') {clickHandler = handleSquareClick; // explicit square handler}
```

Root Cause B — Square Polygon Not Mercator-Corrected

The square used the same raw delta for both latitude and longitude. At non-equatorial latitudes (e.g., Seattle at 47°N), 1° of longitude is ~0.68times shorter than 1° of latitude, producing a visibly distorted rectangle. Additionally, downstream `turf.area()` calls returned incorrect values.

BEFORE (broken):

```
const latDelta = distanceMiles / 69;const lngDelta = distanceMiles / 69; // leftarrow same as latDelta rightarrow distorted at high latitudes
```

AFTER (fixed):

```
const latDelta = distanceMiles / 69;const lngDelta = distanceMiles / (69 * Math.cos(centerLat * Math.PI / 180)); // Mercator correction
```

Root Cause C — Square Polygon Ring Not Closed (Missing 5th Point)

The square was output as 4 coordinate pairs. GeoJSON requires polygon rings to be closed — the last point must repeat the first. `turf.booleanPointInPolygon` threw "ring is not closed" on every property-in-square check.

BEFORE (broken):

```
return [nw, ne, se, sw]; // 4 points — unclosed ring rightarrow turf throws
```

AFTER (fixed):

```
return [nw, ne, se, sw, nw]; // 5 points — closed GeoJSON ring
```

GeoJSON Output Format (All Three Draw Modes)

```
{type: 'Feature',geometry: {type: 'Polygon',coordinates: [[[lng1, lat1], [lng2, lat2], [lng3, lat3], [lng4, lat4], [lng1, lat1]]]},properties: { shapeType: 'square', sideLength: distanceMiles }}
```

Files Affected	MapDrawTool.jsx
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Severity	P1 — Square draw tool completely non-functional
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Bug 3: Sold Date Filter Verification

Architecture Verified — The sold date filter architecture is correct. No code change required.
Ensure the month calculation uses $/(1000 * 60 * 60 * 24 * 30.44)$.

Verification Result

The sold date filter architecture is correct. generateOptimizedRoutes receives a pre-filtered property array from Home.jsx. The filter is applied before calling generateOptimizedRoutes.

Correct implementation (Home.jsx):

```
// Home.jsx — correct wiringconst filteredProps = properties.filter(p => {const monthsAgo =
(Date.now() - new Date(p.sold_date).getTime()) / (1000 * 60 * 60 * 24 * 30.44);return monthsAgo
<= soldMonths;});generateOptimizedRoutes(filteredProps, housesPerRoute, startLocation, allLogs,
options);
```

Correct Data Flow

RouteBuilderSettings.jsx → soldMonths → Home.jsx generateRoutes callback
→ local filter using $/(1000 * 60 * 60 * 24 * 30.44)$ → months, not days → generateOptimizedRoutes(filteredProps, ...)

Critical: The divisor must be 30.44 (average days per month), not 30 or 1 (days). Using $/(1000 * 60 * 60 * 24)$ would compare against days, not months, causing the filter to behave as if soldMonths were a day count.

Status	Architecture verified correct. No code change required.
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Testing Checklist

Bug 1 — Route Generation

Table 4 — Bug 1 QA Test Cases

#	Test	Expected Result
1	Open app, navigate to Route Builder	No crash on load
2	Select territory with 10+ properties, click GENERATE ROUTES	Route displays on map, no TypeError
3	Verify route lines on map	Lines do not cross back over themselves
4	Test with properties that have missing street_name	Appended at end of route, no crash
5	Test with non-numeric house_number like "123A"	No crash, property placed on even side
6	Test with territory containing only 1 property	Single-stop route returned
7	Test with returnToStart: true option	No crash, return distance added to total
8	Open Apple Maps URL from route	Valid URL opens Apple Maps with waypoints
9	Test with all properties having invalid coordinates	Returns empty route gracefully

Bug 2 — Square Draw Tool

Table 5 — Bug 2 QA Test Cases

#	Test	Expected Result
1	Select square draw tool, tap map once	Center marker placed, no crash
2	Tap map second time to set edge	Square polygon drawn, no TypeError
3	Inspect square visually	Looks square (not a distorted rectangle)
4	Verify properties inside square are filtered	Correct properties highlighted
5	Very small tap (accidental touch, <50m)	Ignored — no polygon drawn

6	Test at high latitude (Seattle ~47°N)	Square still looks visually square
7	Switch between circle, polygon, square modes	No crash on mode switch
8	Draw circle, then polygon, then square	All three modes produce valid GeoJSON

Bug 3 — Sold Date Filter

Table 6 — Bug 3 QA Test Cases

#	Test	Expected Result
1	Set sold date filter to 1 month	Only properties sold in last ~30 days appear
2	Set sold date filter to 12 months	All properties sold in last year appear
3	Property sold 35 days ago, filter = 2 months	Property appears (35 days < 2 times 30.44 days)
4	Property sold 35 days ago, filter = 1 month	Property does NOT appear (35 days > 30.44 days)
5	Verify divisor in source	Confirm / (1000 * 60 * 60 * 24 * 30.44) not / (1000 * 60 * 60 * 24)